BANK LOAN ANALYSIS

DATA ANALYTICS PROJECT REPORT

Analyzing loan applications, funding activity, borrower characteristics and repayment performance

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Tools: Python • Pandas • NumPy • Matplotlib • Seaborn • Plotly • Jupyter Notebook

1. Executive Summary

This project examines a financial loan dataset to understand the bank's lending activity, the characteristics of borrowers, the amount of money disbursed, and repayment outcomes. The analysis uses Python to summarize loan applications, calculate key performance indicators (KPIs), classify loan outcomes, and compare funding across time and borrower segments.

The notebook contains 38,576 loan records and 24 columns. It reports total funded loans of $435.76 million and an average interest rate of 12.05%. December 2021 is the latest issue month in the data, with 4,314 applications and $53.98 million funded. Based on the notebook's stated classification, 33,243 applications (86.18%) are good loans (Fully Paid or Current), while 5,333 applications (13.82%) are charged-off loans.

These figures provide a useful overview, but some notebook calculations and labels require correction before the results are treated as final. In particular, the overall amount-received output displays the funded amount by mistake, and the DTI output is labelled as an interest rate. The report flags these issues rather than presenting uncertain values as verified results.

2. Introduction

Banks issue loans to individuals for purposes such as debt consolidation, home improvement, education and other personal financial needs. Lending generates interest income, but it also creates credit risk when borrowers fail to repay according to the agreed terms. Monitoring loan applications, disbursements and repayment outcomes helps a lender understand portfolio activity and identify areas that may need further review.

Bank Loan Analysis applies descriptive data analysis to historical loan records. It focuses on what happened in the dataset: how many loans were recorded, how much was funded, what repayment statuses were reported, and how lending was distributed by month, state, loan term, employment length, loan purpose and home ownership.

3. Business Problem

The business problem is to develop a clear, measurable view of loan applications and repayment performance. Without consistent metrics, it is difficult to compare lending activity over time, understand where funds are being distributed, or assess the balance between performing and charged-off loans.

The project addresses the following questions:

- How many loan applications were recorded, including the latest-month application count?
- What was the total funded amount, and how much was funded in the latest issue month?
- What repayment amount was received from borrowers?
- What are the average interest rate and debt-to-income ratio?
- What proportion of applications are classified as good loans and charged-off loans?
- How does funded amount vary by month, state, loan term, employment length, purpose and home ownership?

4. Project Objectives

- Measure loan application volume and funded amounts.
- Review repayment amounts and compare them with disbursements.

- Classify loans by the status definitions used in the dataset.
- Explore monthly lending trends and regional distribution.
- Compare funding across borrower and loan categories.
- Present results in a structured way that supports further business investigation.

5. Dataset Overview

The notebook loads an Excel file named financial_loan.xlsx. The dataset contains 38,576 rows and 24 columns. Each row represents a loan record. The available fields cover borrower details, loan characteristics, dates and repayment information.

Data group	Example fields	Analytical use
Borrower profile	address_state, emp_length, home_ownership, annual_income, dti	Compare lending across borrower groups.
Loan characteristics	loan_amount, term, int_rate, grade, sub_grade, purpose, installment	Understand loan size, pricing and purpose.
Dates	issue_date, last_payment_date, next_payment_date, last_credit_pull_date	Analyze timing and monthly trends.
Loan outcomes	loan_status, total_payment	Classify repayment status and summarize amounts received.

The issue dates in the notebook range from January 2021 to December 2021. The latest issue date falls in December 2021, so the notebook uses December 2021 as its latest-month period.

6. Tools and Technologies

Tool	Purpose in the project
Python	Main programming language for data analysis.
Pandas	Reading Excel data, filtering, grouping and aggregating records.
NumPy	Numerical operations and calculations.
Matplotlib	Creating line, area, bar and pie-style charts.
Seaborn	Statistical visualization support.
Plotly Express	Creating an interactive treemap.
Jupyter Notebook	Writing, running and documenting the analysis.

7. Methodology

Data loading: The Excel dataset is imported into a Pandas DataFrame.

Data inspection: The notebook reviews the first and last records, dataset dimensions, data types and descriptive statistics.

KPI calculation: Counts, sums and averages are calculated for application volume, funded amount, interest rate and DTI.

Loan classification: Fully Paid and Current records are grouped as good loans; Charged Off records are grouped as bad loans.

Trend analysis: Issue dates are grouped by month to summarize application counts, funded amounts and payments received.

Segment analysis: Funding is grouped by state, loan term, employment length, loan purpose and home ownership.

Visualization: Charts are used to communicate the results and make comparisons easier.

8. Key Performance Indicators

KPI	Notebook result	Meaning
Total loan applications	38,576	Number of loan records in the dataset.
Latest-month applications	4,314	Applications with issue dates in December 2021.
Total funded amount	$435.76 million	Sum of loan_amount across the dataset.
Latest-month funded amount	$53.98 million	Sum of loan_amount for December 2021.
Average interest rate	12.05%	Mean of the int_rate field, multiplied by 100 in the notebook.
Average DTI	13.33% reported	Mean of dti multiplied by 100; the notebook output label is incorrect and the scale should be validated.

The notebook also calculates total_amount_received from the total_payment column. However, the display statement mistakenly formats total_funded_amount instead of total_amount_received. Therefore, the overall amount received is not reliably reported by the current notebook output and should be corrected before publication.

9. Good Loan and Bad Loan Analysis

For this project, good loans are defined as records with a loan_status of Fully Paid or Current. Bad loans are defined as records with a loan_status of Charged Off. This is the classification used in the notebook and should be stated whenever these figures are presented.

Metric	Good loans	Bad loans
Application count	33,243	5,333
Share of all applications	86.18%	13.82%
Funded amount	$370.22 million	$65.53 million
Amount received (as reported for each group)	$435.79 million	$37.28 million

The good-loan share is based on application counts, not on the proportion of money recovered. Also, the Current category represents loans that are still active; it should not be described as fully repaid. The amounts received for the good- and bad-loan groups are notebook outputs and should be checked against the raw total_payment field as part of final validation.

10. Trend and Segment Analysis

10.1 Monthly trends

The notebook groups records by issue month and prepares three monthly views: application count, funded amount and total payment received. These charts can reveal changes in loan demand, disbursement activity and incoming payments over the 2021 period. December 2021 records 4,314 applications and

$53.98 million in funded loans. Monthly patterns should be interpreted as descriptive trends, not proof of seasonality or a cause of changes.

10.2 Regional analysis

Funded amounts are grouped by address_state and displayed as a horizontal bar chart. This view helps identify states with larger or smaller funding totals. A high funded amount does not by itself indicate stronger repayment performance; repayment status should be compared separately.

10.3 Loan term

The notebook compares total funded amount across loan terms using a donut-style chart. This helps show how the portfolio is distributed across repayment periods. The project should retain the exact term categories from the dataset when reporting the chart.

10.4 Employment length

Funding is grouped by employment-length category. This allows comparison of disbursements among borrowers with different reported work histories. It does not establish that employment length causes a particular repayment outcome.

10.5 Loan purpose

The purpose analysis groups funded amounts by the stated reason for borrowing. It can help show which purposes account for larger portions of lending activity and where borrower demand is concentrated.

10.6 Home ownership

A treemap groups funded amounts by home-ownership category. It provides a compact view of how funding is distributed among categories such as RENT and MORTGAGE. These categories should be treated as descriptive borrower attributes, not as standalone measures of creditworthiness.

11. Key Findings

- The dataset contains 38,576 loan records across 24 columns.
- Total funded amount reported by the notebook is $435.76 million.
- December 2021 is the latest issue month and contains 4,314 applications.
- The notebook reports $53.98 million funded in December 2021.
- The average interest rate reported is 12.05%.
- Under the notebook’s classification, 86.18% of applications are good loans and 13.82% are charged off.
- The notebook provides segment views by state, term, employment length, purpose and home ownership.

12. Recommendations

1. Correct the overall amount-received calculation so the displayed value uses total_amount_received rather than total_funded_amount.
2. Correct the DTI output label and confirm whether the dti field is stored as a decimal or percentage before multiplying by 100.

3. Standardize currency notation. The notebook uses dollar symbols in KPI outputs but rupee symbols in some chart labels; confirm the dataset currency and use one unit consistently.
4. Compare charged-off rates by state, loan purpose, term and other segments, not just funded amounts.
5. Review repayment amounts alongside loan status and loan age so active loans are not incorrectly treated as completed repayments.
6. Validate missing values, duplicate records, date formats and unusual values before drawing business conclusions.
7. Add clear chart titles, units, date ranges and definitions to make the analysis understandable to reviewers.

13. Limitations

- The dataset covers issue dates in 2021, so the findings describe that period and may not represent current lending conditions.
- The analysis is descriptive and does not establish causal relationships.
- The current notebook contains calculation and labeling issues that affect the reliability of some displayed KPIs.
- The monthly and segment charts are included in the notebook, but this report does not invent additional monthly or category values that are not printed in the notebook output.
- The dataset currency should be confirmed because the notebook mixes dollar and rupee labels.

14. Conclusion

The Bank Loan Analysis project demonstrates how Python-based data analytics can be used to summarize lending activity, examine repayment outcomes and explore borrower and loan segments. The analysis provides a useful starting point for understanding application volume, funding patterns and portfolio quality. The reported results show 38,576 applications, $435.76 million in funded loans, and a good-loan application share of 86.18% under the project's stated definition.

Before using the report for business decisions or a formal portfolio presentation, the overall amount-received calculation, DTI labeling and currency notation must be corrected and validated. With these checks completed, the project can serve as a clear portfolio example of data cleaning, KPI development, exploratory analysis and business-focused visualization.

Appendix: Core Metric Definitions

Metric	Suggested definition
Total applications	Count of unique loan IDs, after confirming whether each row represents one unique loan.
Total funded amount	Sum of loan_amount.
Total amount received	Sum of total_payment.
Average interest rate	Mean of int_rate, with its stored scale confirmed.
Average DTI	Mean of dti, with its stored scale confirmed.
Good-loan share	(Fully Paid + Current applications) ÷ total applications × 100.
Charged-off share	Charged Off applications ÷ total applications × 100.